



Machine Learning

AI-Driven GRC Auditor

Automated Compliance Solutions

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What is GRC?



GOVERNANCE

*Managing the
company correctly*



RISK

*Finding and fixing
dangers (like
hacking)*



COMPLIANCE

*Following laws and
bank rules*

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AI-Driven GRC
Auditor

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The Problem?

Manual compliance checks are slow, error-prone, and costly. Organizations must review thousands of pages of policies manually, consuming valuable time and resources. Human auditors struggle to keep pace with evolving regulations, leading to gaps and vulnerabilities.

Our goal

Automate compliance auditing using AI to enhance accuracy, reduce costs, and accelerate the audit process.



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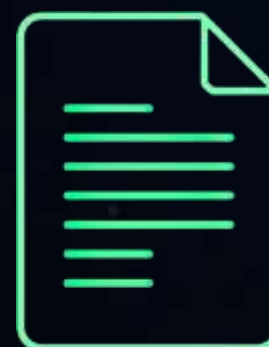


The Role of NLP



TEXT PROCESSING

NLP helps the computer "read" the PDF files just like a human auditor does.



TEXT CHUNKING

Large documents are broken into smaller, meaningful pieces (chunks) for better analysis.



SEMANTIC UNDERSTANDING

Instead of searching for words, NLP understands the "meaning" of the sentences

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The Role of Machine Learning



VECTORIZATION

Converts text into numerical data using advanced embedding models, transforming words into mathematical vectors for AI processing.



PATTERN RECOGNITION

Finds relationships between policies and regulations, identifying connections and similarities across documents.



AI REASONING

Evaluates compliance status and identifies gaps using intelligent decision-making algorithms.

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System Workflow Overview

- 1. Input:** *User uploads a PDF policy.*
- 2. Process:** *AI reads, converts to math, and searches for rules.*
- 3. Output:** *A dashboard showing exactly where the security gaps are.*

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1. Text Ingestion

- **Reading the Document** *The system uses NLP libraries to extract raw text from complex PDF layouts.*
- **Recursive Splitting:** *We split the text into 800-word blocks.*
- **Overlap:** *We keep 100 words common between blocks so no information is lost.*

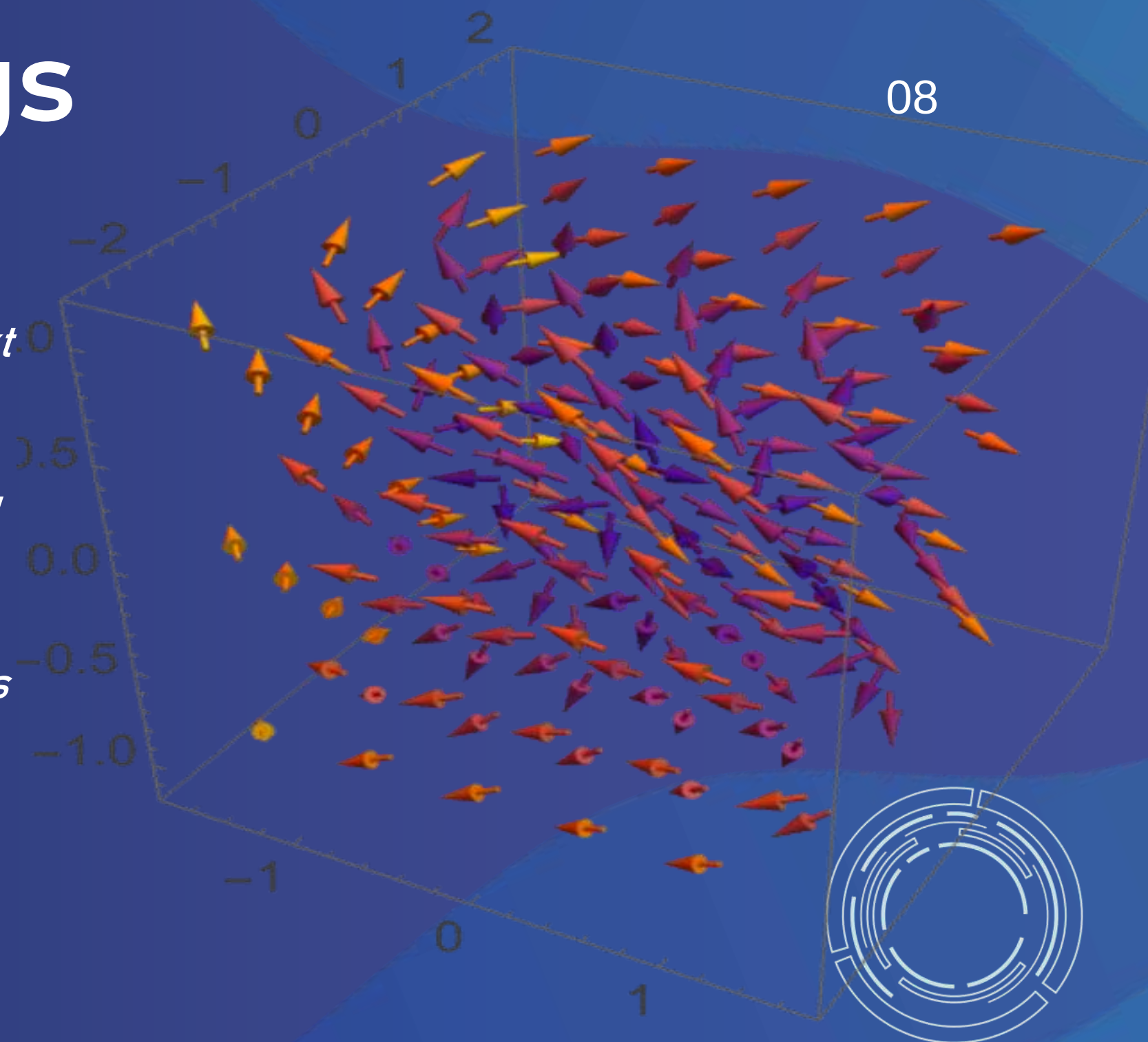
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2. Vector Embeddings

- **Words to Numbers:** *This step changes regular text into numbers called "vectors".*
- **The Nomic Model:** *We use a specific AI tool called the Nomic Model to do this math.*
- **Digital Map:** *Words are placed on a 3D digital map as coordinate points.*



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Semantic Retrieval

➤ **Finding Matches**

When we ask a question, the ML model looks for chunks of text that are mathematically similar to our question.

➤ **KNN Algorithm:** *Finds the "Nearest" neighbors in data.*

➤ **Result:** *It finds the exact page and paragraph related to the audit rule.*



AI Reasoning

We use DeepSeek-R1, a Large Language Model (LLM), to think like an expert auditor.

- *It reads the found text and decides:*
- *Is the bank following the rule?*
- *What is missing?*
- *How dangerous is this mistake? (High/Low Severity)*



Compliance Mapping



Compliance Gap Found	ISO Control	Severity
Shared admin accounts found in IT policy	A.9 (Access Control)	High
Backup files are not encrypted at rest	A.10 (Encryption)	High
Log retention period is only 30 days	A.12 (Logging)	Medium
Incident response team contact missing	A.16 (Incident)	Low



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Questions?